



TEMASEK JUNIOR COLLEGE
PRELIMINARY EXAMINATION
JC2 2021

CANDIDATE NAME

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CIVICS GROUP

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H2 BIOLOGY

Paper 2

Structured Questions

9744/02

25 August 2021

2 hours

Candidates answer on the Question Paper

No Additional Materials are required.

READ THESE INSTRUCTIONS FIRST

Write your name and civics group in the spaces at the top of this page.

Write in dark blue or black pen.

You may use an HB pencil for any diagrams or graphs.

Do not use staples, paper clips, glue or correction fluid.

Answer **all** questions in the spaces provided on the Question Paper.

The use of an approved scientific calculator is expected, where appropriate.

You may lose marks if you do not show your working or if you do not use appropriate units.

The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's use	
1	/ 9
2	/ 10
3	/ 10
4	/ 10
5	/ 10
6	/ 10
7	/ 10
8	/ 11
9	/ 10
10	/ 5
11	/ 5
Total	/ 100

Answer **all** the questions in this section.

- 1** The tertiary structure in a protein refers to the folding of the protein into a precise, compact, globular three-dimensional shape. This shape is held in place by four types of interactions that occur between R groups of amino acid residues.

(a) State the four types of interactions that hold the tertiary structure in shape.

.....
 [1]

- (b)** The Creutzfeldt-Jacob disease (CJD) is a fatal degenerative brain disorder caused by misfolded proteins known as prions. Prions can exist in multiple forms. Examples include the normal prion protein (PrP^{C}) and the disease-causing prion protein (PrP^{Sc}) which causes CJD.

Fig. 1.1 below shows the structures of PrP^{C} and PrP^{Sc} respectively.

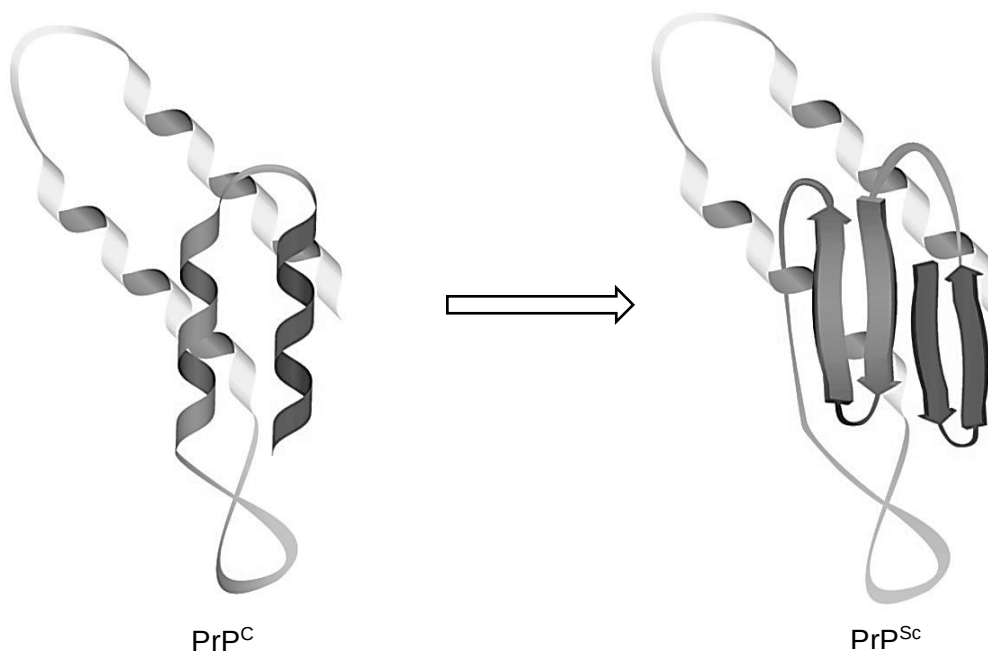


Fig. 1.1

It is noticed that in CJD, each PrP^{C} may misfold into PrP^{Sc} , which then accumulate in large numbers and aggregate (i.e. come together), resulting in the typical sponge-like structure of the brain.

- (i)** With reference to Fig. 1.1 and the information given above, suggest how CJD may arise.

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 [2]

- (ii) Suggest why prions challenge the concept of the levels of protein structure.

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 [1]

Although CJD is classified as a disease, scientists have noted that there is little to no immune response amongst diagnosed patients.

- (iii) Suggest why there is a lack of immune response in patients with CJD.

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 [1]

- (c) (i) Contrast between the structures of PrP^{Sc} and haemoglobin.

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 [2]

Non-covalent interactions are present in haemoglobin, and this allows for cooperative oxygen binding, thereby increasing the efficiency of oxygen transport.

- (ii) Explain how cooperative oxygen binding increases the efficiency of oxygen transport.

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 [2]

[Total: 9]

- 2 Fig. 2.1 is an electron micrograph of the process of DNA replication in a eukaryotic cell. The arrows point to replication bubbles. Each replication bubble arises from one origin of replication in the DNA.

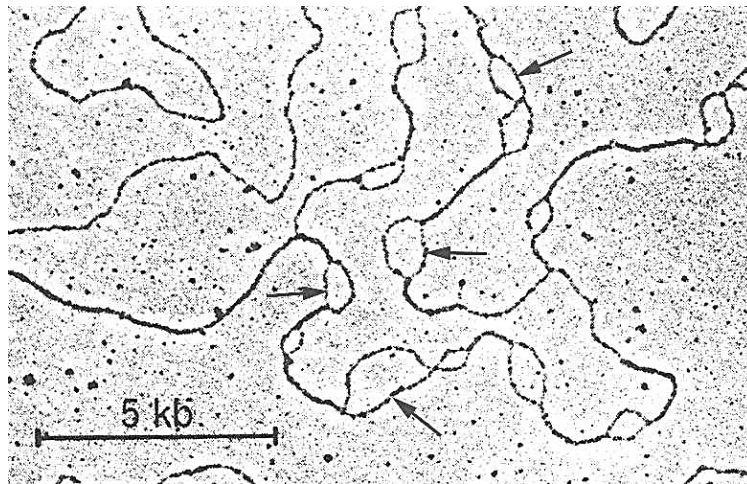


Fig. 2.1

- (a) (i) Explain why eukaryotic cells have multiple origins of replication.

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 [1]

- (ii) Explain how the replication bubble is formed and its significance to the semi-conservative mode of DNA replication.

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 [3]

- (b) Fig. 2.2 shows part of a DNA undergoing replication.

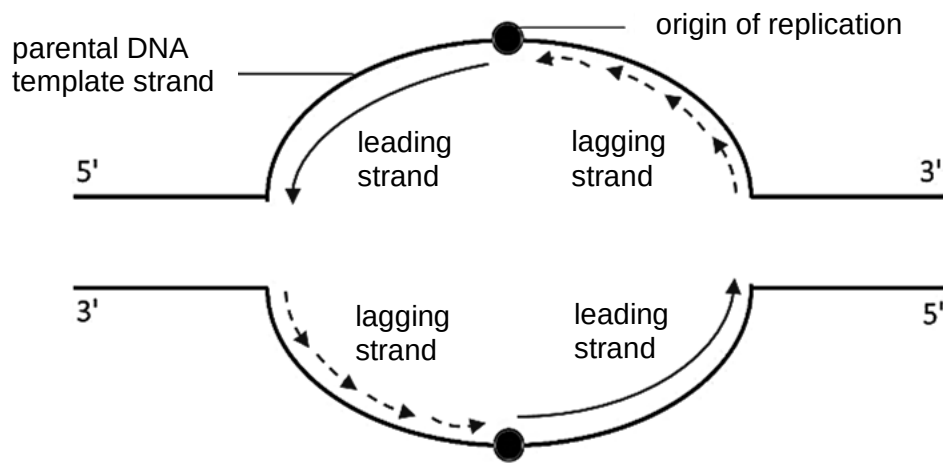


Fig. 2.2

- (i) Circle the end(s) of the DNA template(s) on Fig. 2.2 where the end replication problem would be experienced.

[1]

- (ii) Explain the reason for the end replication problem.

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..... [2]

- (c) Contrast DNA replication and polymerase chain reaction.

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..... [3]

[Total: 10]

- 3 The 1968 flu pandemic is one of the deadliest pandemics in history, killing approximately 4 million individuals globally. This pandemic was caused by the novel H3N2 influenza virus.
- (a) Fig. 3.1 shows how the H3N2 influenza virus attacks human respiratory epithelial cells inside the nose.

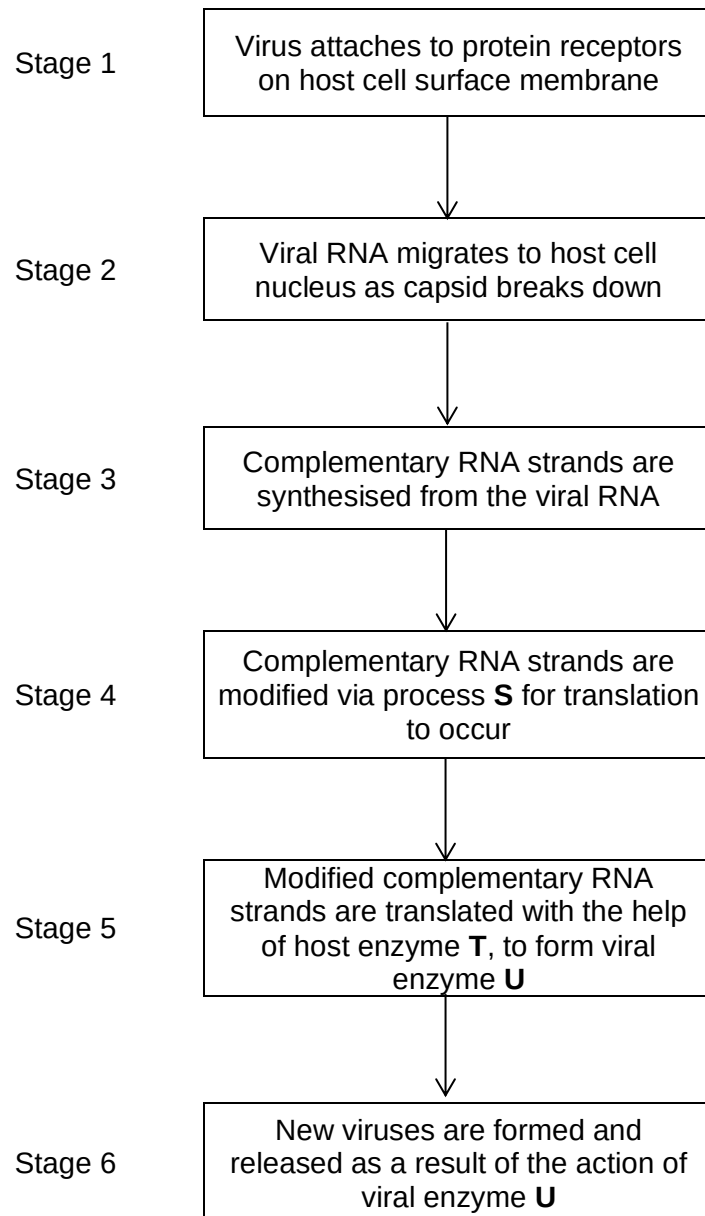


Fig. 3.1

- (i) Explain why the H3N2 influenza virus is only able to attack human respiratory epithelial cells.

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 [1]

- (ii) Suggest a specific identity of process **S**, and explain why process **S** is required for translation to occur.

identity
 explanation

 [2]

- (iii) Suggest the identity of host enzyme **T**, and explain how host enzyme **T** helps with the translation of the modified complementary RNA strands.

identity
 explanation

 [2]

- (iv) Explain how viral enzyme **U** helps with the release of new influenza viruses.

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 [1]

(b) The novel H3N2 influenza virus arose via a mechanism shown in Fig. 3.2.

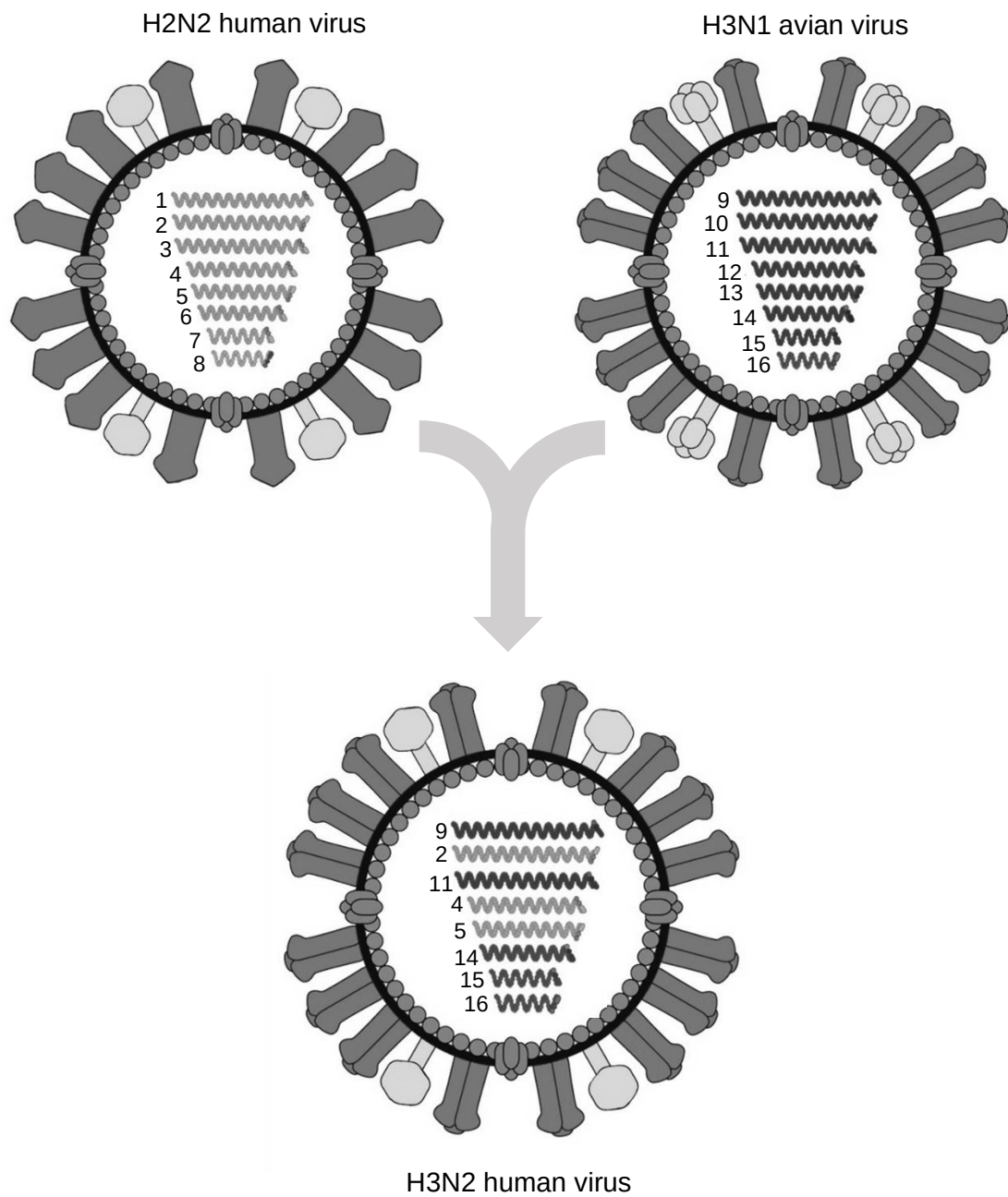


Fig. 3.2

- (i) With reference to Fig. 3.2, describe how the novel H3N2 influenza virus arose.

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..... [3]

- (ii) Suggest why the human population was susceptible to the novel H3N2 influenza virus.

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..... [1]

[Total: 10]

- 4 Meiosis, also known as reduction division, is a process by which a parent cell divides to produce gametes.

- (a) Assuming that the total number of chromosomes in the parent cell is $2n = 46$, complete Table 4.1.

Table 4.1

stage	number of chromosomes per cell	ploidy	amount of DNA
start of interphase	46	$2n$	$2x$
end of interphase			
end of meiosis I (after cytokinesis)			
end of meiosis II (after cytokinesis)	23	n	x

[2]

- (b) Fig. 4.1 shows the length of kinetochore spindle fibres during different stages of meiosis.

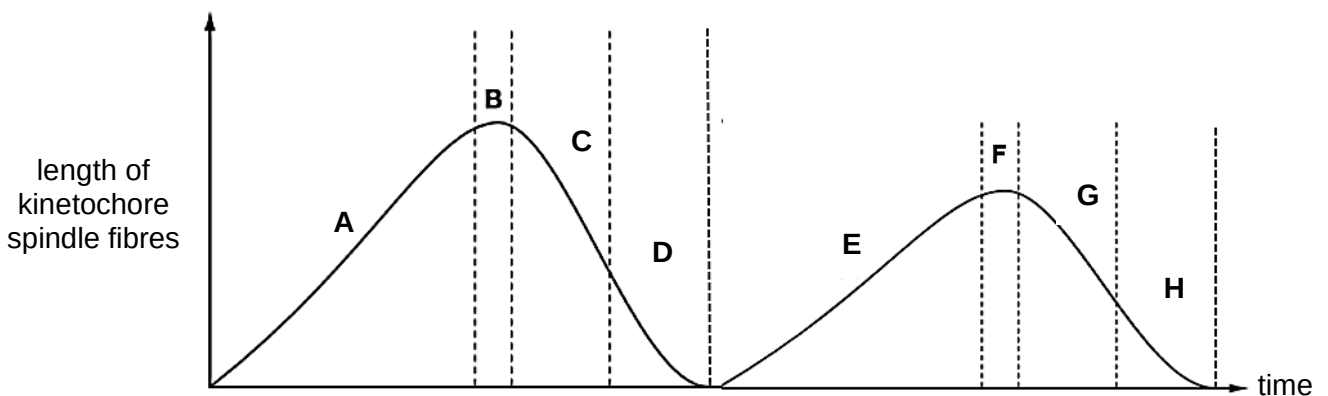


Fig. 4.1

- (i) Explain the changes in the length of the kinetochore spindle fibres from A to D.

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[3]

Fig. 4.2 shows a particular stage in meiosis.



Fig. 4.2

- (ii) State the region in Fig. 4.1 that corresponds to the stage shown in Fig. 4.2.

..... [1]

- (iii) Identify region(s) in Fig. 4.1 that contribute(s) to genetic variation in gametes, and explain how the genetic variation is brought about.

region(s)

explanation

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..... [2]

Colchicine is a drug that inhibits spindle fibre assembly. It is commonly used by farmers to induce polyploidy in plants, and this is a common technique used to overcome the sterility of a hybrid species (hybrid sterility) during plant breeding.

For example, triticale is the hybrid of wheat ($2n = 28$) and rye ($2n = 14$). The initial hybrids are sterile as they have an odd number of chromosomes ($2n = 21$). Upon addition of colchicine, the hybrids become fertile, and they can be propagated to become triticale.

- (b) Explain how the addition of colchicine overcomes hybrid sterility in triticale.

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..... [2]

[Total: 10]

- 5 The majority of lung cancers are caused by long term exposure to several classes of carcinogens present in cigarette smoke. However, there are cases of lung cancers arising in the absence of detectable cigarette exposure.

Analysis of different gene mutations in individuals who smoked ('ever smokers') and those who did not ('never smokers') suggests that lung cancers in never smokers may follow a very different molecular pathway of malignant transformation.

Fig. 5.1 shows the percentage of individuals in US and Asia with lung cancer having mutations in the proto-oncogenes, epidermal growth factor receptor (*EGFR*) gene and *Ras* gene.

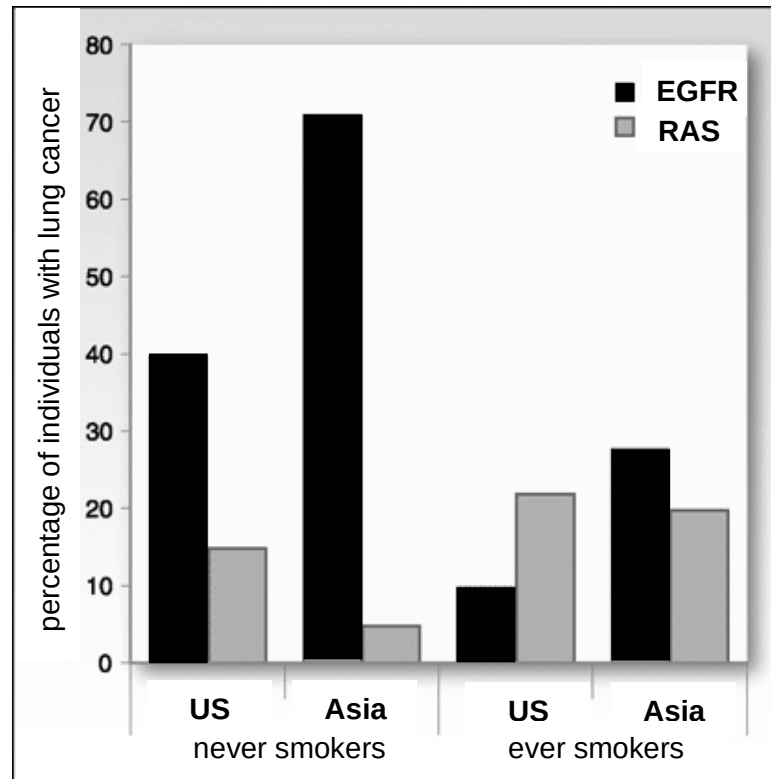


Fig. 5.1

With reference to Fig. 5.1,

- (a) (i)** Describe the effect of the two gene mutations on the occurrence of cancer in Asians.

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..... [2]

- (ii)** With reference to Fig. 5.1 and your biological knowledge, explain how never smokers in US and Asia developed lung cancer.

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..... [3]

Cancers are traditionally viewed as genetic disorders, but it has become increasingly evident that aberrant epigenetic modifications also play major roles in cancer development. Epigenetics is the study of heritable changes in gene expression that occur without changes in DNA sequences.

Fig 5.2 shows the major epigenetic changes in lung cancer.

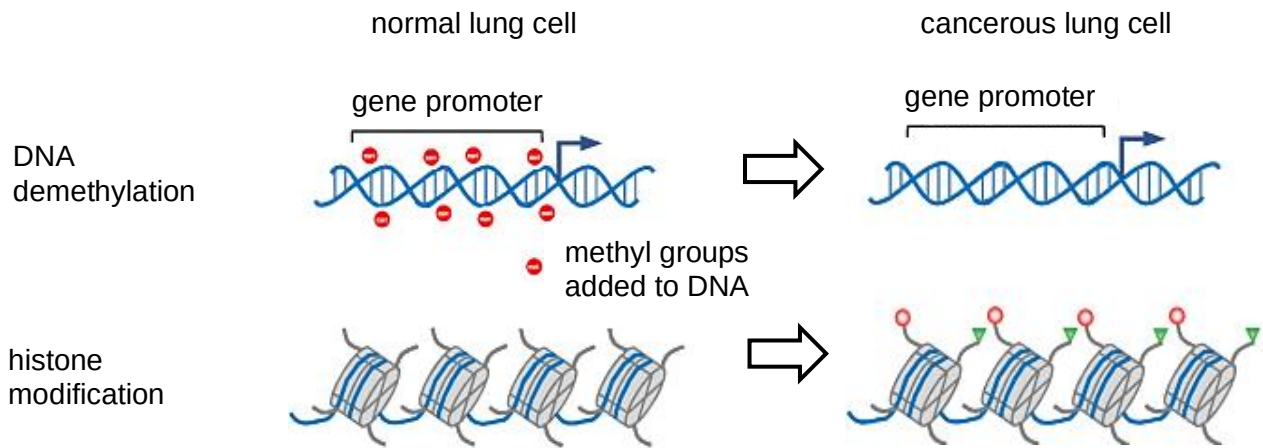


Fig. 5.2

(b) (i) Explain how the epigenetic changes shown in Fig. 5.2 may result in higher gene expression.

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A non-coding RNA (ncRNA) is a single stranded RNA molecule that is transcribed from DNA but not translated into proteins.

(ii) Suggest how ncRNA may be used as a drug to treat lung cancers arising from epigenetic changes.

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..... [1]

[Total: 10]

- 6 In humans, a gene on chromosome 19 is responsible for the formation of a glycolipid found in red blood cell surface membranes. This glycolipid is known as antigen H. The gene has two alleles, H and h. The product of allele h is non-functional.

A second gene on chromosome 9 codes for an enzyme that modifies antigen H. There are three alleles of this gene:

- I^A which results in the conversion of antigen H to antigen A
- I^B which results in the conversion of antigen H to antigen B
- i which does not produce a functional product so does not modify antigen H.

There are four possible phenotypes depending on the combination of the alleles of these two genes:

- phenotype A, where individuals have antigen A only
- phenotype B, where individuals have antigen B only
- phenotype AB, where individuals have both antigen A and antigen B
- phenotype O, where individuals have neither antigen A nor antigen B.

Two parents, both with phenotype AB, have a child with phenotype O.

- (a) State the genotype of both parents for these **two** genes.

parent 1

parent 2

[1]

- (b) The same parents then have a second child.

Use a genetic diagram to show the possible genotypes and corresponding phenotypes of the second child.

[5]

The interpupillary distance (IPD) is the distance in millimetres between the centres of the pupils of the eyes. Fig. 6.1 shows how IPD is measured.

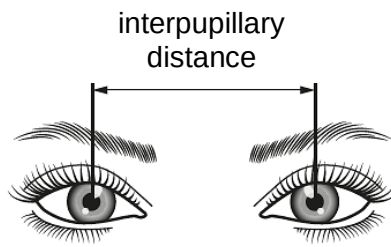


Fig. 6.1

IPD is one example of a characteristic of human facial structure that shows variation.

Fig. 6.2 shows the pattern of variation in IPD in a large sample of adults.

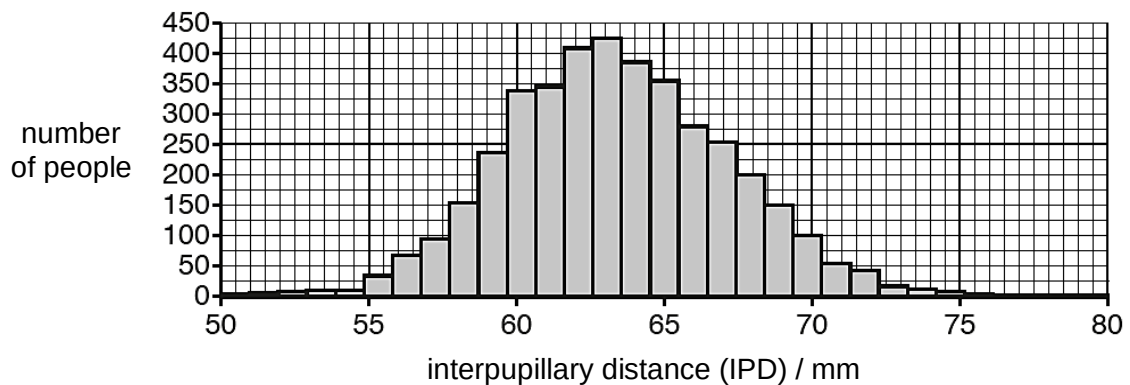


Fig. 6.2

- (a) (i) Name the type of variation shown in Fig. 6.2.

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- (ii) Suggest and explain how genes and the environment contribute to variation in IPD in humans.

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[Total: 10]

- 7 Cytochrome c is an electron carrier involved in oxidative phosphorylation. It is a protein that transfers electrons between two large proteins found in the electron transport chain. It removes electrons from cytochrome bc1 and transfers them to cytochrome oxidase, as shown in Fig. 7.1.

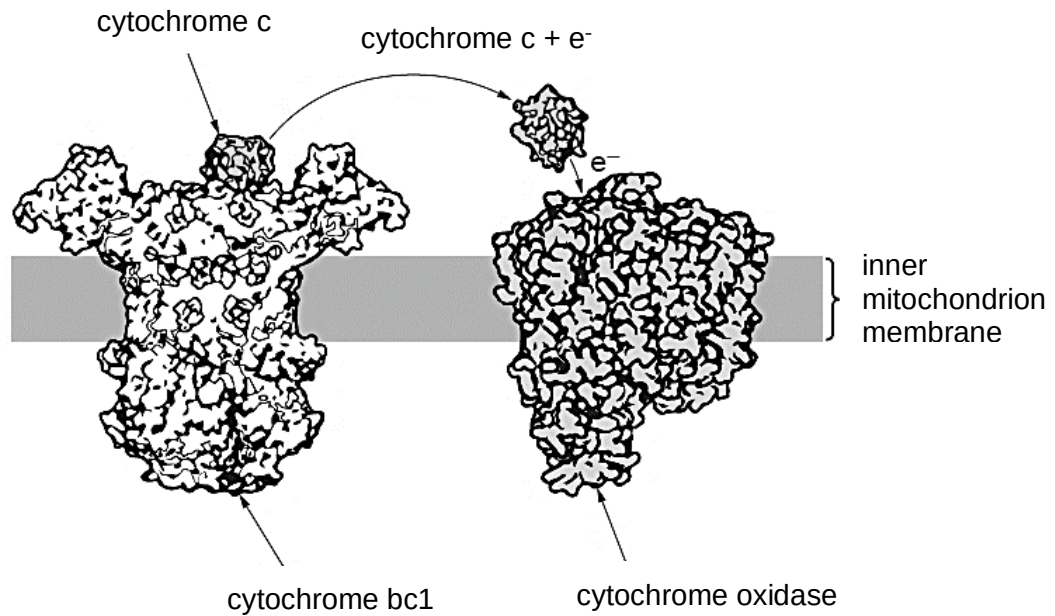


Fig. 7.1

- (a) Cytochrome oxidase is the final protein in the electron transport chain. It is an enzyme which transfers the electrons to an oxygen molecule.

State the final product of this reaction in the mitochondrion.

..... [1]

- (b) The cytochrome oxidase molecule is different in different species. Fig. 7.2 shows the enzyme from a cow and a bacterium. The shaded section in each represents the polypeptide chains which are very similar in both organisms. This is the area of the enzyme which binds to the cytochrome c.

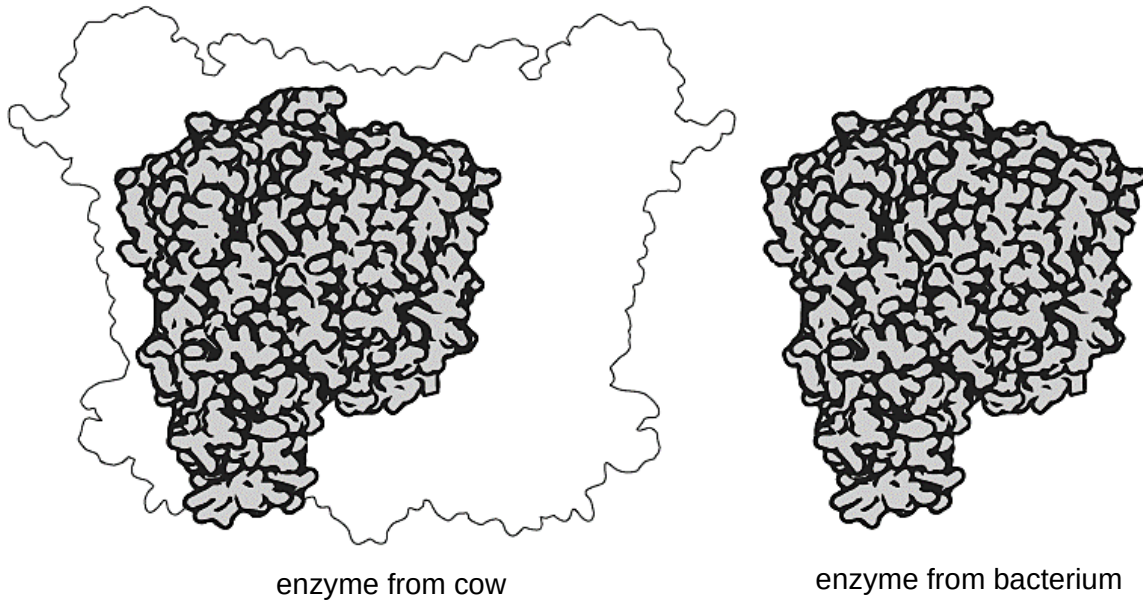


Fig. 7.2

- (iii) Suggest why the shaded regions are similar in all organisms.

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 [1]

- (iv) Suggest why cytochrome c detaches from the cytochrome oxidase molecule when the electrons it carries are removed.

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 [2]

- (c) Cytochrome bc1 is a proton pump that transfers protons into the inter-membrane space.

Antimycin is a chemical that binds to cytochrome bc1 and blocks its action.

Explain what happens to the pH of the inter-membrane space when antimycin is present.

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..... [2]

- (d) Oligomycin A is another chemical which acts by inhibiting protons from passing through ATP synthase.

- (i) Explain why the presence of oligomycin A in a human causes high levels of reduced NAD to form in the cell.

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- (ii) Suggest why lactate begins to accumulate in the blood if high levels of reduced NAD form.

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..... [2]

[Total: 10]

- 8 An investigation was carried out into the effect of eating two different sources of carbohydrate on the concentration of glucose and the concentration of insulin in the blood plasma. The carbohydrate sources used were glucose and rice. The carbohydrate in rice is mainly in the form of starch.

Volunteers were fed either 50g of glucose or the quantity of rice known to contain 50g of carbohydrate. The concentration of glucose and insulin in their blood plasma was then measured at intervals for 3 hours. The results are shown in Fig. 8.1 and 8.2.

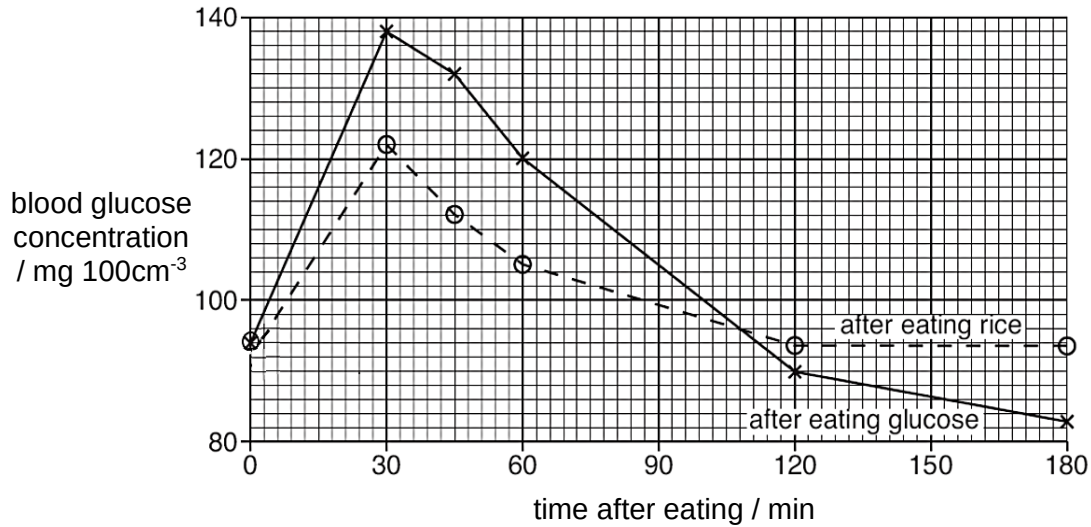


Fig. 8.1

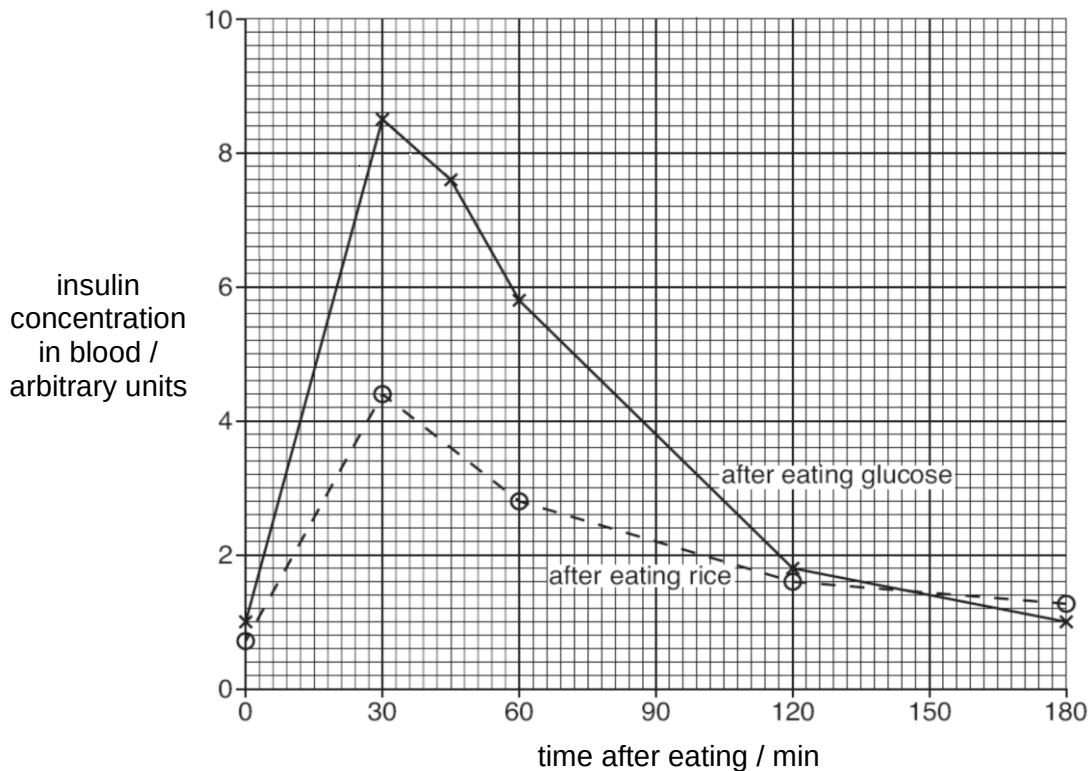


Fig. 8.2

- (a) (i) Calculate the difference between the maximum concentrations of glucose in the blood in the first 30 minutes after eating rice and after eating glucose.

Present your answer as a percentage of the blood glucose concentration at the start of the experiment. Show your working.

[1]

- (ii) Give an explanation for the difference you have calculated in (a)(i).

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 [1]

- (b) (i) With reference to Fig. 8.1 and Fig. 8.2, explain the difference between the concentration of insulin in the blood in the first 30 minutes after eating glucose and eating rice.

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 [2]

- (ii) Explain why the insulin in the blood starts to decrease after 30 minutes.

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 [1]

- (c) Using your biological knowledge on cell signaling, explain how insulin decrease blood glucose concentration.

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..... [4]

- (d) Explain the advantages of using stem cells to treat patients with end-stage liver disease where their diseased liver cells are unable to carry out normal functions.

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..... [2]

[Total: 11]



- 9 Modern horses are a group of mammals adapted to living in grassland regions. Their evolution can be reconstructed from fossil records.

Table 9.1 shows the evolution of the teeth and height of the modern horse over 50 million years.

Table 9.1

Name	Teeth	Height / cm	Environmental conditions		
<i>Equus</i> (2 mya – today)	“Herbivorous teeth”	150	Predators present.	Grassland region. Mostly fibrous grass.	Edible insects extinct.
<i>Meryohippus</i> (26 mya)	Long cubical molars and flat grinding surfaces	100	Predators present.	Evolution of grass. Forest started to recede.	Edible insects becoming extinct.
<i>Mesohippus</i> (35 mya)	“Omnivorous teeth”	60	Predators present.	Forest filled with trees and shrubs (above 60cm).	Edible insects found on the ground and upper part of the shrubs.
<i>Eohippus</i> (50 mya)	Short, bumpy molars	25	Predators present.	Forest filled with trees and shrubs (above 60cm).	Edible insects found on the ground.

mya - million years ago

- (a) With reference to Table 9.1, explain how natural selection led to the evolution of the horse over time.

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..... [5]

- (b) The evolution of the horse provides examples of both micro-evolution and macro-evolution. Explain the link between micro-evolution and macro-evolution.

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Horses have hooved feet and an odd number of toes on each foot. This feature supports the common ancestry of horses and tapirs. Unlike the horse, the tapirs retained their jungle specialisation.

- (c) Suggest how this difference led to the speciation of their common ancestors into modern horses and tapirs starting from about 26 mya.

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..... [2]

[Total: 10]



- 10 Countries that have a high number of cases of malaria also have problems with diseases caused by bacteria. This means that many people in these countries are prescribed antibiotics, such as penicillin, for the treatment of bacterial infections.

(a) Outline how penicillin acts on bacterial cells.

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..... [2]

The female *Anopheles* mosquito is the vector of the *Plasmodium* (a type of unicellular eukaryotic organism) pathogen that causes malaria. The insect takes in *Plasmodium* when taking a blood meal from an infected person. At a later stage, the insect can transmit the pathogen when taking a blood meal from an uninfected person.

(b) Research has shown that *Plasmodium* is not always transmitted to uninfected people.

Two main reasons for this have been suggested.

- The immune system of mosquitoes kills *Plasmodium* while it is still in their gut.
- Bacteria living in the gut of mosquitoes compete with *Plasmodium* so it does not survive to continue its life cycle.

A study was carried out to see if taking antibiotics affects the risk of transmission of malaria.

Some observations and results of the study are summarised in Fig. 10.1.

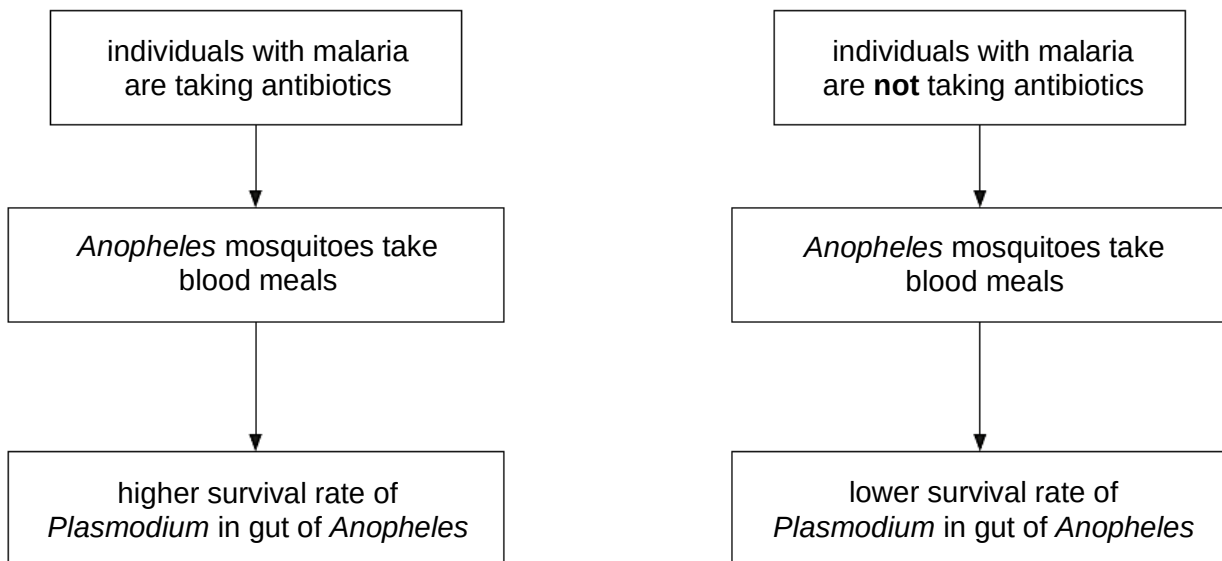


Fig. 10.1

Suggest explanations for the results shown in Fig. 10.1 and comment on the importance of these results for doctors working in countries that have malaria and a high number of bacterial infections.

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..... [3]

[Total: 5]



- 11** Fig. 11.1 shows the percentage of a population with HIV and the number of new tuberculosis (TB) infections from 1980 to 2010.

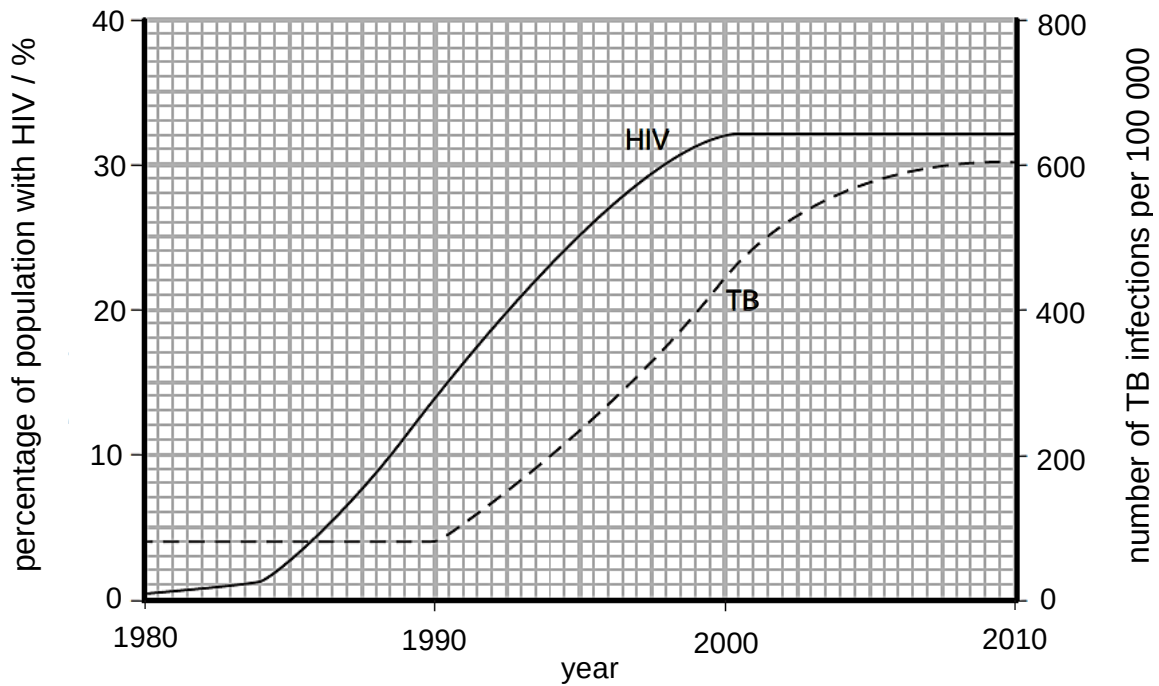


Fig. 11.1

- (a)** Describe the evidence shown in this graph that suggests a relationship between infection with HIV and TB infection.

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..... [2]

TB is caused when droplets, containing the bacterium *Mycobacterium tuberculosis*, are inhaled into the lungs. In the lungs, large numbers of the bacterium are formed rapidly. These can be ingested by macrophages. Eventually, tubercles (tissue masses), containing dormant bacteria inside macrophages, may form.

(b) Describe the events from the ingestion of *M. tuberculosis* to the formation of tubercles.

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..... [3]

[Total: 5]

